

ADEBOWALE RICHARD OYELOWO

Backend Engineer | Python, FastAPI, PostgreSQL & Systems

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SUMMARY

Backend engineer building and deploying Python services with FastAPI, Flask, and PostgreSQL. Built multi-tenant SaaS systems, authentication infrastructure, and production services end to end, from schema design and API implementation through testing, CI/CD, and deployment. Strong interest in systems engineering, networking, and reliable backend architecture.

SKILLS

Languages: Python, C, SQL, JavaScript/TypeScript

Backend: FastAPI, Flask, REST APIs, asynchronous services, API design

Data: PostgreSQL, SQLAlchemy, asyncpg, Alembic, Redis

Security: JWT, OAuth2, RBAC, bcrypt, multi-tenant architecture

Infrastructure: Docker, GitHub Actions, CI/CD, Railway, Render

Testing: pytest, pytest-asyncio, HTTPX, integration testing, TDD

EXPERIENCE

Freelance Backend Engineer

January 2025 - Present

Remote, solo

- Built and deployed production Python applications covering API design, database schemas, service layers, authentication, testing, and CI/CD.
- Designed multi-tenant RBAC with centralized ORM query scoping, ensuring database queries are restricted to the authenticated tenant.
- Built CI/CD pipelines with GitHub Actions that run the test suite and block deployments when tests fail.
- Debugged production issues by tracing failures to underlying data and concurrency behavior rather than applying application-level workarounds.

SELECTED PROJECTS

Clustra, Multi-Tenant Project Management API (FastAPI, asyncpg, SQLAlchemy)

- Built an asynchronous API with FastAPI, asyncpg, and SQLAlchemy, with database migrations applied automatically before the application accepts traffic.
- Designed multi-tenant RBAC with centralized query scoping so database queries are restricted to the authenticated organization.
- Covered asynchronous service logic with pytest-asyncio and configured CI to prevent releases when tests fail.

Snip, Containerized URL Shortener (FastAPI, PostgreSQL, Docker)

- Built and deployed a containerized URL shortener using Docker, PostgreSQL, multi-stage builds, and health checks.
- Diagnosed a concurrency issue in short-code generation and replaced the problematic approach with an atomic counter-backed Base62 scheme, eliminating collision retries and application-level locking.
- Added integration tests covering redirects, short-code resolution, and administrative flows; tests caught four bugs before production deployment.

DevBrain, Flask Quiz Platform (Flask, SQLAlchemy, Jinja2, PostgreSQL)

- Built a full-stack quiz platform with Flask, SQLAlchemy, and Jinja2, including user-facing quiz flows and an administrative content-management interface.
- Designed database-backed content with draft/published states and JSON import tooling for managing quiz content.
- Added 36 tests covering application behavior, authentication, content management, and core quiz functionality.

Business Dashboard, Data Management & Analytics Application (Flask, SQLAlchemy, Chart.js)

- Built a Flask application for customer and order management with database-backed CRUD operations and dashboard visualizations.
- Added CSV and JSON bulk-import workflows for customer and order data.
- Enforced per-user data isolation at the query layer so users only access records associated with their account.

OPEN SOURCE

gatevault, Python Authentication Library (PyPI)

- Authored and published a framework-agnostic authentication library providing JWT lifecycle management, bcrypt password hashing, OAuth2, and decorator-based route protection.
- Centralized token issuance and expiry management through a TokenManager using dependency injection.
- Added pytest coverage for token creation, decoding, expiration, and signature-tampering cases; automated package releases with GitHub Actions.

EDUCATION

Harvard CS50x & CS50P, Computer Science & Python Engineering, via edX

2025